

1.0 INTRODUCTION

On behalf of the City of Huntington Beach, EEC Environmental (EEC) has prepared this Site Assessment Report for the Cameron Ln. Property located at 17631 Cameron Lane, Huntington Beach, California (subject property; Figure 1, *Site Location Map*). The purpose of the site assessment was to address findings of EEC's Phase I Environmental Site Assessment (ESA) dated March 18, 2020, which identified potential use of pesticides/herbicides and lead-based paint in connection with former agricultural use and residential use as environmental concerns and recommended soil sampling. The Phase I ESA and the site assessment activities were completed in connection with proposed redevelopment of the subject property.

This report documents the methods, procedures, and results of the site assessment activities at the subject property. All work was completed in general accordance with EEC's April 2, 2020 proposal to the City of Huntington Beach (EEC, 2020) and subsequent correspondence between the City of Huntington Beach and EEC. All work was performed under the supervision of a California licensed Professional Geologist (PG) and the Orange County Health Care Agency (OCHCA).

1.1 Scope of Work

The completed site assessment activities included the following tasks:

- Preparing a site-specific health and safety plan (HASP) appropriate for the scope of work.
- Contacting Underground Service Alert (USA) 72 hours prior to initiation of fieldwork.
- Advancing 35 borings for the collection of soil and groundwater samples from depths ranging from approximately 0.5 feet below ground surface (bgs) to 24 feet bgs.
- The collection and analysis of 86 soil samples for organochlorine pesticides (OCPs), herbicides, lead, Title 22 Metals, and hexavalent chromium and 3 groundwater samples for hexavalent chromium.
- Excavation of soil containing elevated concentrations of lead and OCPs at two locations.

2.0 BACKGROUND

2.1 Site Description

The approximately 0.79-acre subject property consists of a residential lot identified as Assessor Parcel Number (APN) 167-042-08, 17631 Cameron Lane in Huntington Beach, Orange County, California. The east side of the subject property is currently developed with an unoccupied single-family residence that was developed in 1947. A Phase I ESA was completed for the subject property on March 18, 2020 (EEC, 2020). The Phase I ESA indicated that based on a review of historical aerial photographs, the subject property appeared to be used for agricultural purposes (row crops) from the 1930s to the 1950s and given this usage, it is possible that pesticides and/or herbicides were once used onsite. Additionally, due to the date of construction of the residence, lead-based paint may have been used on the subject property residence. Based on this information, the Phase I ESA recommended soil sampling to assess for potential residual pesticides and herbicides from former agricultural use and to assess for potential lead in the location of the current building exterior.

2.2 Physiography

The most recent topographic map coverage of the subject property is provided by the U.S. Geological Survey (USGS) 7.5-minute, Newport Beach, California quadrangle map, dated 2012 (Figure 1). According to the USGS topographic map, the subject property is located at an elevation of approximately 25 to 30 feet above mean sea level (msl). The topography in the general subject property area slopes gently to the north-northeast.

2.3 Geology and Hydrogeology

Onsite activities described in this report indicated that soil at the subject property consists primarily of silt at 1 foot bgs. At five feet bgs, soil consists primarily of clayey sand and sandy clay. At 8 feet bgs soil consists of silt, clayey silt, silty sand, sand, sandy clay, and clayey sand. At 20 feet bgs, soil consists of silt, clayey sand, and clayey silt. At 24 feet bgs, the maximum depth explored, soil consists of sand and silty sand.

According to a Quarterly Status Report, Second Quarter 2019 for a site located approximately 500 feet northwest of the subject property, the groundwater flow direction is southwest (Atlas Environmental Engineering, Inc., 2019). During onsite activities, groundwater was encountered at 23 and 24 feet bgs.

3.0 SITE INVESTIGATION ACTIVITIES

Between April 6 and May 11, 2020, a total of 35 soil borings were advanced to depths ranging from approximately 0.5 to 8 feet bgs for the collection of soil samples at the subject property. In addition, three groundwater samples were collected and two soil excavations were completed.

3.1 Pre-Field Activities

Prior to beginning soil sampling activities, EEC completed the following tasks necessary to initiate field activity.

3.1.1 Health and Safety Plan

To protect workers, EEC prepared a HASP as appropriate for the scope of work. The HASP informed all site workers of potential risks during field sampling activities and provided engineering controls to avoid physical and chemical hazards. The HASP addressed potential physical and chemical hazards associated with field operations. The HASP outlined the steps to prevent exposures from all site activities. The HASP also described the safe use of equipment, evaluation of hazards, safe investigation procedures, personal protective equipment, emergency procedures and training required to conduct the site assessment under typical Hazardous Waste Operations and Emergency Response protocol.

3.1.2 Utility Clearance

EEC contacted USA and provided the specific property boundaries so that the proposed work area was provided to utility owners responding to the USA ticket. USA is a regional notification center that notifies owners and operators of subsurface utilities (water, gas, electric, sewer, oil lines, etc.) and informs them

of a contractor's intent to perform subsurface work. EEC notified USA of the intent to perform subsurface work a minimum of 72 hours prior to the beginning of the on-site activities.

3.2 Hand-Auger Soil Boring Advancement and Soil Sampling

Between April 6 and April 28, 2020, EEC advanced a total of 27 soil borings at the subject property (B1 through B18, B4-A, B6-A, and B26-B32; Figure 2, *Boring Location Map*) to total depths of 3 feet bgs. The borings were advanced using a hand auger. Once each sampling depth was reached, the hand auger was removed from the borehole and an undisturbed soil sample was collected using a slide hammer sampler including a stainless steel sampling sleeve. Following sample retrieval, both ends of the sleeve were covered with a Teflon sheet and sealed with a plastic end cap. Soil samples were then labeled with an identification number, sealed in a plastic bag, and placed into a chilled cooler.

An unused portion of soil from each sampling interval was used for field screening using a photoionization detector (PID) for evaluation of organic vapor content and for logging lithology. Field screening was conducted by placing a portion of the unpreserved soil in a sealable plastic bag for approximately 10 minutes. The organic vapor concentration in the headspace of the plastic bag was then measured using a PID. Soil descriptions and PID measurements were recorded and documented on field notes. All field activities were performed under the direct oversight of a California PG.

All soil samples were collected and transferred under proper chain-of-custody protocols to Eurofins Calscience (Calscience) of Garden Grove, a California state-certified laboratory. Selected soil samples were analyzed for organochlorine pesticides (OCPs) by United States Environmental Protection Agency (USEPA) Method 8081A, herbicides by USEPA Method 8151A, California Title 22 metals by USEPA Method 6010B, mercury by USEPA Method 7471A, and hexavalent chromium by USEPA Method 7199.

The specific soil sampling method for each sample location is provided in Table 3-1, *Soil Sampling Matrix*, on Page 6 of this report. Because multiple phases of soil sampling and analysis were completed, Section 3.2.1 below provides a timeline summary to clarify the sequence of events that took place between April 6 and April 28, 2020. Specific details related to analytical results, screening levels, and soil excavations are provided in Sections 4.1 and 5.1 of this report.

3.2.1 Summary of Phases of Hand-Auger Soil Sampling

The initial sampling event took place at the subject property on April 6, 2020. The purpose of this initial sampling event was to determine if residual pesticides or lead were present in soil as a result of former agricultural use and potential use of lead-based paint on the residence as identified in EEC's Phase I ESA dated March 18, 2020. Six hand auger borings (B1 through B6) were advanced and soil samples were collected from 0.5 and 3 feet bgs. The soil samples were submitted to Calscience and analyzed for OCPs and lead by USEPA Methods 8081A and 6010B.

Based on the results of the soil sampling, which identified elevated concentrations of lead and OCPs in B4 and B6, respectively, four additional hand-auger borings (B7 through B10) were advanced on April 13, 2020 and soil samples were collected from 0.5 and 3 feet bgs for additional site coverage. The soil samples were submitted to Calscience and analyzed for OCPs and lead by USEPA Methods 8081A and 6010B. Results of the April 13, 2020 soil sampling indicated that elevated concentrations of OCPs were found in B7, on the northwest portion of the subject property.

Due to the expeditious nature of the project, on April 13, 2020, soil at the location of borings B4 and B6 was excavated and soil samples were collected from confirmation borings at the sidewalls of the excavations (B15 through B18 around B4; B11 through B14 around B6) and submitted to Calscience for analysis of OCPs and lead by USEPA Methods 8081A and 6010B. Elevated concentrations of OCPs and lead were not found in confirmation borings around B4 and B6, with the exception of boring B11, which had elevated concentrations of OCPs. Approximately 1.39 cubic yards of material was removed from the location of boring B4 and approximately 1.39 cubic yards of material was removed from the location of boring B6.

Based on elevated concentrations of OCPs in B7 and B11, step-out hand-auger borings were advanced adjacent to B7 (B29 through B32) and B11 (B26 through B28) on April 16, 2020. Soil samples were collected from 0.5 and 3 feet bgs and submitted to Calscience for analysis of OCPs by USEPA Method 8081A. The only elevated concentrations of pesticides detected in the step-out borings were in B29 and B32, adjacent to boring location B7. The following additional tasks were planned to address the elevated concentrations of OCPs, but were not completed due to scope changes that occurred during the course of the project: excavation around B11 and excavation and confirmation soil sampling around B7, B29, and B32.

On April 27, 2020, the City of Huntington Beach entered into a Remedial Action Agreement with the OCHCA. The OCHCA indicated that select soil samples should also be analyzed for Title 22 Metals, mercury, hexavalent chromium, and herbicides. On April 29 and 30, 2020, soil samples from B4 and B6 through B10 were analyzed for California Title 22 metals by USEPA Method 6010B, mercury by USEPA Method 7471A, and hexavalent chromium by USEPA Method 7199 and soil samples from B7, B9, and B10 were analyzed for herbicides by USEPA Method 8151A. Because the laboratory hold time was exceeded, soil from B4 and B6 could not be analyzed for herbicides. Therefore, on April 28, 2020, hand-auger soil borings B4-A and B6-A were advanced near B4 and B6. Soil samples were collected from 0.5 and 3 feet bgs and submitted to Calscience for analysis of herbicides by USEPA Method 8151A.

3.3 Geoprobe Drilling and Soil and Groundwater Sampling

Results of the additional soil analyses completed on April 29 and 30, 2020 indicated that elevated concentrations of hexavalent chromium were found in multiple soil samples at 0.5 and 3 feet bgs. In order to better understand the lateral and vertical distribution of hexavalent chromium at the subject property, and to determine if groundwater was impacted, EEC oversaw Strongarm Environmental Field Services, Inc. (Strongarm) drill a total of 8 borings for the collection of soil and groundwater samples on May 11, 2020. The borings were placed in a grid pattern laterally over the subject property and close to previous boring locations; therefore, they were given similar nomenclature (B1-A, B4-B, B5-A, B6-B, B7-A, B8-A, B9-A, and B10-A; Figure 2).

The borings were advanced using Geoprobe direct-push drilling technology and hand-auger equipment. Direct-push drilling rigs are hydraulically powered soil probing machines that use both static force and

percussion to drive steel boring rods into the subsurface. Before the initiation of drilling, Strongarm cleared the upper 5 feet of the borehole with a hand auger to verify the absence of underground utilities or other shallow obstructions.

A soil sample was collected from 4 feet bgs from each boring using the hand auger. The hand auger was removed from the borehole and an undisturbed soil sample was collected using a slide hammer sampler including a stainless steel sampling sleeve. Following sample retrieval, both ends of the sleeve were covered with a Teflon sheet and sealed with a plastic end cap. During direct-push drilling, continuous soil samples were collected using 5-foot long acetate sleeves. The direct-push rig hydraulically drove the sampling sleeve the entire 5-foot interval. Following sample retrieval the portion of the sleeve chosen for samples (i.e., 5 feet bgs, 6 feet bgs, 7 feet bgs, and 8 feet bgs) was cut and the ends of each sample were covered with Teflon sheets and sealed with plastic end caps. Soil samples were then labeled with an identification number, sealed in a plastic bag, and placed into a chilled cooler.

An unused portion of soil from each sampling interval was used for field screening using a PID for evaluation of organic vapor content and for logging lithology. Field screening was conducted by placing a portion of the unpreserved soil in a sealable plastic bag for approximately 10 minutes. The organic vapor concentration in the headspace of the plastic bag was then measured using a PID. Soil descriptions and PID measurements were recorded and documented on field notes. All field activities were performed under the direct oversight of a California PG.

The collected soil samples were transferred under proper chain-of-custody protocols to Calscience and analyzed for hexavalent chromium by USEPA Method 7199, as shown on Table 3-1 on Page 6 of this report.

3.3.1 Groundwater Sampling

Following soil sampling, groundwater samples were collected from three borings (B4-B, B8-A, and B10-A). Groundwater samples were collected with a Hydropunch-type sampling device. The rod was withdrawn to expose the screen of the sampling device and a stainless steel bailer was then lowered into the borehole to collect a groundwater sample. Samples were labeled, sealed in a plastic bag, and placed into a chilled cooler. Collected groundwater samples were submitted to Calscience for analysis of hexavalent chromium by USEPA Method 218.6.

Table 3-1, Soil Sampling Matrix

Location/Rationale	Boring IDs	Depth of Borings (bgs)	Date Sampled	Soil Sampling	
				Sample Depth	Analytes
Residence exterior/initial sampling	B1-B3	3 feet	April 6, 2020	0.5 and 3 feet	Lead, OCPs
Residence exterior/initial sampling	B4	3 feet	April 6, 2020	0.5 and 3 feet	Lead, OCPs, Title 22 Metals, Mercury, Hexavalent Chromium
Undeveloped area/initial sampling	B5	3 feet	April 6, 2020	0.5 and 3 feet	Lead, OCPs
Undeveloped area/initial sampling	B6	3 feet	April 6, 2020	0.5 and 3 feet	Lead, OCPs, Title 22 Metals, Mercury, Hexavalent Chromium
Northwest corner/additional coverage	B7	3 feet	April 13, 2020	0.5 and 3 feet	Lead, OCPs, Herbicides, Title 22 Metals, Mercury, Hexavalent Chromium
North center/additional coverage	B8	3 feet	April 13, 2020	0.5 and 3 feet	Lead, OCPs, Title 22 Metals, Mercury, Hexavalent Chromium
Northeast and southwest corners/additional coverage	B9, B10	3 feet	April 13, 2020	0.5 and 3 feet	Lead, OCPs, Herbicides, Title 22 Metals, Mercury, Hexavalent Chromium
Adjacent to B6/confirmation borings	B11-B14	3 feet	April 13, 2020	0.5 and 3 feet	Lead, OCPs
Adjacent to B4/confirmation borings	B15-B18	3 feet	April 13, 2020	0.5 and 3 feet	Lead, OCPs
Adjacent to B11/step-out borings	B26-B28	3 feet	April 16, 2020	0.5 and 3 feet	OCPs
Adjacent to B7/step-out borings	B29-B32	3 feet	April 16, 2020	0.5 and 3 feet	OCPs
Adjacent to B4/herbicide sampling	B4A	3 feet	April 28, 2020	0.5 and 3 feet	Herbicides
Adjacent to B6/herbicide sampling	B6A	3 feet	April 28, 2020	0.5 and 3 feet	Herbicides
Adjacent to B1 and B10/hexavalent chromium sampling	B1-A and B10-A	8 feet	May 11, 2020	4 feet	Hexavalent chromium
Adjacent to B4, B5, B6, B7, B8, and B9/hexavalent chromium sampling	B4-B, B5-A, B6-B, B7-A, B8-A, and B9-A,	8 feet	May 11, 2020	4, 5, 6, 7, and 8 feet	Hexavalent chromium

Key:

bgs = below ground surface

OCPs = organochlorine pesticides

4.0 SAMPLING RESULTS

4.1 Soil

A total of 86 soil samples were collected and analyzed for OCPs, herbicides, California Title 22 metals, lead, mercury, and hexavalent chromium between April 6 and May 11, 2020 (Table 3-1). The soil laboratory results are summarized below and in Table 1, *Summary of Soil Analytical Results – Metals* and Table 2, *Summary of Soil Analytical Results – Pesticides and Herbicides*. Estimated values (J), detected below the reporting limit and above the laboratory method detection limit are included in the summary. Analytical results were compared to the USEPA Regional Screening Levels (RSLs) for residential soil and DTSC Screening Levels (SLs) for residential soil (Cleanup Standards). If both RSLs and SLs were available for a particular analyte, then the more stringent value was used. Laboratory analytical results, chain-of-custody documentation, and quality control data are provided in Appendix A, *Laboratory Analytical Reports, Chain-of-Custody Record, and QA/QC Data*.

4.1.1 Metals

- With the exception of lead and hexavalent chromium, metals detected in soil samples were at concentrations below their respective Cleanup Standard, or established background levels.
- Lead was detected at 101 milligrams per kilogram (mg/kg) in B4-0.5, exceeding the Cleanup Standards. The deeper sample (3 feet bgs) collected from the same boring contained lead at a concentration of 2.81 mg/kg, below Cleanup Standards; therefore, vertical delineation was achieved. Lead was not detected above Cleanup Standards in confirmation samples collected after excavation of soil around B4 (Section 5.1).
- Hexavalent chromium was detected throughout the subject property at depths ranging from 0.5 feet bgs to 8 feet bgs. Concentrations ranged from 200 to 980 micrograms per kilogram (µg/kg), some of which exceeded Cleanup Standards.

4.1.2 Pesticides and Herbicides

- Several organochlorine pesticides were detected in shallow soil, including 4,4'-Dichlorodiphenyldichloroethane (4,4'-DDD), 4,4'-Dichlorodiphenyldichloroethylene (4,4'-DDE), 4,4'-Dichlorodiphenyltrichloroethane (4,4'-DDT), alpha-chlordane, chlordane, dieldrin, and toxaphene. All concentrations were below Cleanup Standards for residential soil, with the exception of toxaphene and 4,4'-DDE. No other pesticides were detected above Cleanup Standards.
- Toxaphene was detected in three samples (B6-0.5, B7-0.5, and B10-0.5) at concentrations ranging between 500 µg/kg and 1,600 µg/kg, which exceeded the USEPA RSL of 490 µg/kg and DTSC SL of 450 µg/kg. 4,4'-DDE was detected in one sample (B7-0.5) at 3,100 µg/kg, which exceeded the USEPA RSL of 2,000 µg/kg (DTSC SL not established).
- Herbicides were not detected above laboratory reporting limits in any of the samples that were analyzed (B4-A, B6-A, B7, B9, and B10 at 0.5 and 3 feet bgs).

4.2 Groundwater

On May 11, 2020, groundwater samples were collected from three of the geoprobe boring locations (B8-A/GW-1, B4-B/GW-2, and B10-A/GW-3) in order to determine if hexavalent chromium in soil had impacted groundwater. Collected groundwater samples were transferred under proper chain-of-custody protocols to Calscience for analysis of hexavalent chromium by USEPA Method 218.6.

Hexavalent chromium was not detected above the laboratory reporting limit of 0.038 micrograms per liter ($\mu\text{g/L}$) in any of the groundwater samples. It was detected at 0.086 $\mu\text{g/L}$ in the equipment blank sample, which is laboratory-provided water used as a quality control sample.

5.0 INVESTIGATION DERIVED WASTE AND EXCAVATED SOIL

Excavated soil, soil cuttings, and decontamination water generated during this assessment work were stored onsite in ten United States Department of Transportation (DOT) approved 55-gallon drums. The drums were labeled documenting the accumulation start date, description of contents, and applicable contact information. Wastes will be profiled, manifested, and disposed of within 90 days of generation. If available at the time of report preparation, the manifest(s) signed by the disposal facility will be included in the final report.

6.0 CONCLUSIONS AND RECOMMENDATIONS

This site assessment and limited soil excavation was performed to address the findings of the Phase I ESA report. Based on the results of the site assessment and limited excavation activities, EEC has the following conclusions:

- Soil impacted with lead, pesticides, and hexavalent chromium was found at the subject property.
- Lead-impacted soil identified at the location of boring B4 was excavated and no additional investigation or excavation is warranted for lead impacts.
- Pesticide-impacted soil at the location of boring B6 was partially excavated; however, multiple locations of elevated detections of pesticides remain at the subject property.
- Hexavalent chromium was detected throughout the subject property at depths of up to 8 feet bgs; however, because hexavalent chromium was not detected in groundwater it appears to be a limited and isolated concern.

Based on the above conclusions, EEC recommends:

- The ground surface at the subject property be covered with asphalt or concrete to eliminate the risk of dermal contact with soil at the subject property.
- Any soil excavated from short- or long-term redevelopment be managed, handled, and disposed of in accordance with all regulatory requirements to minimize exposure to workers and the community.

- If a change to use of the subject property occurs, the soil contamination must be addressed and additional assessment may also be required.
- The OCHCA must be notified if there are any changes in site use and/or if there are any future findings of additional contamination at the site.

7.0 REFERENCES

Atlas Environmental Engineering, Inc. (2018, May 9). *Quarterly Status Report, 2nd Quarter 2019, OCHCA Case #93UT070, G&M Oil Company Station #35, 17501 Beach Boulevard, Huntington Beach, California 92647.*

EEC Environmental (2020, March 18). *Phase I Environmental Site Assessment, 17631 Cameron Lane and 17642 Beach Boulevard, Huntington Beach, California.*

United States Geological Survey (USGS). 7.5-Minute Topographic Quadrangle Map, Newport Beach, California, 2012

Table 1, Summary of Soil Analytical Results - Metals
17631 Cameron Lane
Huntington Beach, California

Sample ID	Sample Depth (feet bgs)	Date	Antimony (mg/Kg)	Arsenic (mg/Kg)	Barium (mg/Kg)	Beryllium (mg/Kg)	Cadmium (mg/Kg)	Chromium (mg/Kg)	Cobalt (mg/Kg)	Copper (mg/Kg)	Lead (mg/Kg)	Molybdenum (mg/Kg)	Nickel (mg/Kg)	Selenium (mg/Kg)	Silver (µg/Kg)	Thallium (mg/Kg)	Vanadium (µg/Kg)	Zinc (mg/Kg)	Mercury (mg/Kg)	Hexavalent Chromium (µg/Kg)
			USEPA Method 6010B																	USEPA Method 7471A
B1-0.5	0.5	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	44.6	NA	NA	NA	NA	NA	NA	NA	NA	NA
B1-3.0	3.0	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	3.23	NA	NA	NA	NA	NA	NA	NA	NA	NA
B1A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<200
B2-0.5	0.5	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	15.6	NA	NA	NA	NA	NA	NA	NA	NA	NA
B2-3.0	3.0	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	3.52	NA	NA	NA	NA	NA	NA	NA	NA	NA
B3-.05	0.5	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	5.29	NA	NA	NA	NA	NA	NA	NA	NA	NA
B3-3.0	3.0	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	2.57	NA	NA	NA	NA	NA	NA	NA	NA	NA
B4-0.5	0.5	04/6/2020	4.61	10.3	150	0.760	2.11	21.6	9.81	38.0	101	ND<0.260	17.4	ND<0.781	ND<0.260	ND<0.781	40.1	240	0.210	500 J
B4-3.0	3.0	04/6/2020	ND<0.735	2.56	81.1	0.922	ND<0.490	17.3	7.60	6.29	2.81	0.696	12.7	ND<0.735	ND<0.245	ND<0.735	35.9	35.1	ND<0.0794	290 J
B4B-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	350 J
B4B-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	300 J
B4B-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	380 J
B4B-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	330 J
B4B-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	410 J
B5-0.5	0.5	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	17.9	NA	NA	NA	NA	NA	NA	NA	NA	NA
B5-3.0	3.0	04/6/2020	NA	NA	NA	NA	NA	NA	NA	NA	2.43	NA	NA	NA	NA	NA	NA	NA	NA	NA
B5A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	300 J
B5A-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	540
B5A-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	330 J
B5A-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	580
B5A-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	200 J
B6-0.5	0.5	04/6/2020	ND<0.785	8.44	113	1.12	1.00	23.1	10.7	20.3	7.62	0.617	16.7	ND<0.785	ND<0.262	ND<0.785	48.7	75.3	ND<0.0806	290 J
B6-3.0	3.0	04/6/2020	ND<0.769	3.27	81.5	0.858	ND<0.513	14.0	6.43	10.3	12.8	0.712	12.0	ND<0.769	ND<0.256	ND<0.769	31.1	45.2	ND<0.0847	820
B6B-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	570
B6B-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	430
B6B-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	620
B6B-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	350 J
B6B-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<400
B7-0.5	0.5	04/13/2020	1.0	3.67	50.9	ND<0.248	0.748	9.5	3.44	17.8	30.0	0.422	6.41	ND<0.743	0.311	ND<0.743	13.2	97.1	0.245	NA ⁽¹⁾
B7-3.0	3.0	04/13/2020	1.37	ND<0.754	54.1	0.437	ND<0.503	7.48	6.71	6.16	5.16	ND<0.251	6.34	ND<0.754	ND<0.251	ND<0.754	19.8	16.0	0.0817	380 J
B7A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<1,000
B7A-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<400
B7A-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	420
B7A-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	270 J
B7A-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	330 J
USEPA RSLs for Soil - Residential (µg/kg) ⁽²⁾			31	0.68	15,000	1,600	71	120,000	23	3,100	82	390	1,500	390	390	0.78	390	23,000	23	300 ⁽⁴⁾
DTSC SL for Soil - Residential (µg/kg) ⁽³⁾			--	0.11	--	16 ⁽³⁾	910	--	--	--	80 ⁽³⁾	--	820 ⁽³⁾	--	--	--	--	--	1.0 ⁽³⁾	300 ⁽⁴⁾

Table 1, Summary of Soil Analytical Results - Metals
17631 Cameron Lane
Huntington Beach, California

Sample ID	Sample Depth (feet bgs)	Date	Antimony (mg/Kg)	Arsenic (mg/Kg)	Barium (mg/Kg)	Beryllium (mg/Kg)	Cadmium (mg/Kg)	Chromium (mg/Kg)	Cobalt (mg/Kg)	Copper (mg/Kg)	Lead (mg/Kg)	Molybdenum (mg/Kg)	Nickel (mg/Kg)	Selenium (mg/Kg)	Silver (mg/Kg)	Thallium (mg/Kg)	Vanadium (mg/Kg)	Zinc (mg/Kg)	Mercury (mg/Kg)	Hexavalent Chromium (µg/Kg)
			USEPA Method 6010B																	USEPA Method 7471A
B8-0.5	0.5	04/13/2020	0.913	1.83	54.3	0.372	ND<0.513	6.62	4.24	7.55	25	ND<0.256	5.86	ND<0.769	ND<0.256	ND<0.769	17.3	26.4	ND<0.0820	480
B8-3.0	3.0	04/13/2020	1.42	7.30	109	ND<0.253	ND<0.505	3.97	2.40	8.77	2.45	1.53	4.27	ND<0.758	ND<0.253	ND<0.758	9.2	14.6	ND<0.0833	ND<200
B8A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<800
B8A-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<200
B8A-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	250 J
B8A-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	240 J
B8A-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	210 J
B9-0.5	0.5	04/13/2020	ND<0.739	3.15	57.5	0.282	ND<0.493	5.45	3.55	8.26	9.59	ND<0.246	5.12	ND<0.739	ND<0.246	ND<0.739	14.3	22.3	ND<0.0847	ND<200
B9-3.0	3.0	04/13/2020	ND<0.743	2.16	56.5	ND<0.248	ND<0.495	4.23	2.66	4.9	3.33	ND<0.248	3.85	ND<0.743	ND<0.248	ND<0.743	10.9	12.4	ND<0.0862	360 J
B9A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<800
B9A-5.0	5.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	980
B9A-6.0	6.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	320 J
B9A-7.0	7.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<200
B9A-8.0	8.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<200
B10-0.5	0.5	04/13/2020	1.15	5.07	78.3	0.428	ND<0.488	9.51	5.03	18.5	47.3	0.260	12.5	ND<0.732	ND<0.244	ND<0.732	21.0	51.4	ND<0.0862	430
B10-3.0	3.0	04/13/2020	ND<0.725	1.55	40.0	0.378	ND<0.483	6.28	2.55	3.52	2.54	0.654	5.24	ND<0.725	ND<0.242	ND<0.725	22.6	14.2	ND<0.0877	310 J
B10A-4.0	4.0	05/11/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	270
B11-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	10.2	NA	NA	NA	NA	NA	NA	NA	NA	NA
B11-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	2.44	NA	NA	NA	NA	NA	NA	NA	NA	NA
B12-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	19.6	NA	NA	NA	NA	NA	NA	NA	NA	NA
B12-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	4.07	NA	NA	NA	NA	NA	NA	NA	NA	NA
B13-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	18.0	NA	NA	NA	NA	NA	NA	NA	NA	NA
B13-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	4.02	NA	NA	NA	NA	NA	NA	NA	NA	NA
B14-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	4.64	NA	NA	NA	NA	NA	NA	NA	NA	NA
B14-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	3.57	NA	NA	NA	NA	NA	NA	NA	NA	NA
B15-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	11.3	NA	NA	NA	NA	NA	NA	NA	NA	NA
B15-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	10.9	NA	NA	NA	NA	NA	NA	NA	NA	NA
B16-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	9.86	NA	NA	NA	NA	NA	NA	NA	NA	NA
B16-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	5.05	NA	NA	NA	NA	NA	NA	NA	NA	NA
B17-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	4.08	NA	NA	NA	NA	NA	NA	NA	NA	NA
B17-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	3.06	NA	NA	NA	NA	NA	NA	NA	NA	NA
B18-0.5	0.5	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	10.5	NA	NA	NA	NA	NA	NA	NA	NA	NA
B18-3.0	3.0	04/13/2020	NA	NA	NA	NA	NA	NA	NA	NA	11	NA	NA	NA	NA	NA	NA	NA	NA	NA
B31-0.5	0.5	04/16/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<400
B31-3.0	3.0	04/16/2020	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	ND<800
USEPA RSLs for Soil - Residential (µg/Kg) ⁽²⁾			31	0.68*	15,000	1,600	71	120,000	23	3,100	82	390	1,500	390	390	0.78	390	23,000	23	300 ⁽⁴⁾
DTSC SL for Soil - Residential (µg/kg) ⁽³⁾			--	0.11*	--	16 ⁽³⁾	910	--	--	--	80 ⁽³⁾	--	820 ⁽³⁾	--	--	--	--	--	1.0 ⁽³⁾	300 ⁽⁴⁾

Key:

NA - Not analyzed

(a) = Environmental Protection Agency (EPA) Region IX Regional Screening Levels for Residential Soils

bgs = below ground surface

RSLs= Regional Screening Levels

J = Result is less than the reporting limit but greater than or equal to the method detection limit

mg/kg = milligrams per kilogram

ND<X = not detected above the reporting limit or method detection limit

SL = Screening level

USEPA = United States Environmental Protection Agency

µg/kg = micrograms per kilogram

Notes:

(1) Sample was too small for analysis; B31 was selected as an alternate.

B31 was diluted due to the nature of the soil matrix.

(2) DTSC Human and Ecological Risk Office (HERO) Human Health Risk

Assessment (HHRA) Note 3, April 2019.

(3) Value is DTSC SL non-cancer endpoint

(4) RSL and SL were converted to µg/kg

-- = Note 3 SL not available or references USEPA RSLs

Yellow highlighted results exceed RSL and SL

*DTSC documentation indicates the background concentration for arsenic in Southern California is 12 mg/kg

Table 2, Summary of Soil Analytical Results - Pesticides and Herbicides
17631 Cameron Lane
Huntington Beach, California

Sample ID	Sample Depth (feet bgs)	Date	4,4'-DDD (µg/Kg)	4,4'-DDE (µg/Kg)	4,4'-DDT (µg/Kg)	alpha- Chlordane (µg/Kg)	Chlordane (µg/Kg)	Dieldrin (µg/Kg)	Toxaphene (µg/Kg)	Other Pesticides (µg/Kg)	Herbicides (µg/Kg)
			Pesticides - USEPA Method 8081A								USEPA Method 8151A
B1-0.5	0.5	04/6/2020	ND<5.0	10	5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B1-3.0	3.0	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B2-0.5	0.5	04/6/2020	12	360	180	ND<1.0	ND<25	5.0	ND<25	ND	NA
B2-3.0	3.0	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B3-.05	0.5	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B3-3.0	3.0	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B4-0.5	0.5	04/6/2020	ND<5.0	43	22	6.7	33	2.8	ND<25	ND	NA
B4-3.0	3.0	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B4A-0.5	0.5	04/28/2020	NA	NA	NA	NA	NA	NA	NA	NA	ND
B4A-3.0	3.0	04/28/2020	NA	NA	NA	NA	NA	NA	NA	NA	ND
B5-0.5	0.5	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B5-3.0	3.0	04/6/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B6-0.5	0.5	04/6/2020	ND<25	1,400	350	7.2	54	ND<1.0	770	ND	NA
B6-3.0	3.0	04/6/2020	ND<5.0	18	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B6A-0.5	0.5	04/28/2020	NA	NA	NA	NA	NA	NA	NA	NA	ND
B6A-3.0	3.0	04/28/2020	NA	NA	NA	NA	NA	NA	NA	NA	ND
B7-0.5	0.5	04/13/2020	32	3,100	1,300	ND<1.0	ND<25	ND<1.0	1,600	ND	ND
B7-3.0	3.0	04/13/2020	ND<5.0	16	6.6	ND<1.0	ND<25	ND<1.0	ND<25	ND	ND
B8-0.5	0.5	04/13/2020	ND<5.0	270	63	ND<1.0	ND<25	1.5	110	ND	NA
B8-3.0	3.0	04/13/2020	ND<5.0	13	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B9-0.5	0.5	04/13/2020	ND<5.0	17	9.6	ND<1.0	ND<25	ND<1.0	ND<25	ND	ND
B9-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	ND
B10-0.5	0.5	04/13/2020	ND<5.0	770	410	1.8	ND<25	ND<1.0	500	ND	ND
B10-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	ND
B11-0.5	0.5	04/13/2020	13	1,900	610	1.5	ND<25	ND<1.0	1,200	ND	NA
B11-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B12-0.5	0.5	04/13/2020	5.7	240	100	ND<0.99	ND<25	1.6	59	ND	NA
B12-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<0.99	ND<25	ND<0.99	ND<25	ND	NA
B13-0.5	0.5	04/13/2020	16	500	100	ND<1.0	ND<25	2.5	180	ND	NA
B13-3.0	3.0	04/13/2020	ND<5.0	7.7	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B14-0.5	0.5	04/13/2020	8.2	110	53	ND<0.98	ND<25	ND<0.98	ND<25	ND	NA
B14-3.0	3.0	04/13/2020	ND<4.9	ND<4.9	ND<4.9	ND<0.98	ND<25	ND<0.98	ND<25	ND	NA
B15-0.5	0.5	04/13/2020	ND<4.9	ND<4.9	ND<4.9	ND<0.98	ND<25	ND<0.98	ND<25	ND	NA
B15-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<0.99	ND<25	ND<0.99	ND<25	ND	NA
B16-0.5	0.5	04/13/2020	ND<5.0	22	12	2.2	ND<25	ND<1.0	ND<25	ND	NA
B16-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B17-0.5	0.5	04/13/2020	ND<5.0	6.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B17-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B18-0.5	0.5	04/13/2020	ND<4.9	20	8.2	2.1	ND<25	ND<0.99	ND<25	ND	NA
B18-3.0	3.0	04/13/2020	ND<5.0	ND<5.0	ND<5.0	1.0	ND<25	ND<1.0	ND<25	ND	NA
B26-0.5	0.5	04/16/2020	ND<5.0	230	43	ND<1.0	ND<25	1.1	120	ND	NA
B26-3.0	3.0	04/16/2020	ND<5.0	29	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B27-0.5	0.5	04/16/2020	ND<5.0	220	52	ND<1.0	ND<25	ND<1.0	100	ND	NA
B27-3.0	3.0	04/16/2020	ND<5.0	400	77	ND<1.0	ND<25	ND<1.0	230	ND	NA
B28-0.5	0.5	04/16/2020	ND<5.0	420	130	ND<1.0	ND<25	2.6	180	ND	NA
B28-3.0	3.0	04/16/2020	ND<5.0	33	ND<5.0	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B29-0.5	0.5	04/16/2020	95	1,000	360	2.5	ND<25	ND<1.0	940	ND	NA
B29-3.0	3.0	04/16/2020	ND<5.0	61	27	ND<1.0	ND<25	ND<1.0	37	ND	NA
B30-0.5	0.5	04/16/2020	38	460	120	2.4	ND<25	ND<1.0	410	ND	NA
B30-3.0	3.0	04/16/2020	ND<5.0	20	8.2	ND<1.0	ND<25	ND<1.0	ND<25	ND	NA
B31-0.5	0.5	04/16/2020	11	650	130	2.1	ND<25	ND<1.0	380	ND	NA
B31-3.0	3.0	04/16/2020	ND<5.0	190	62	ND<1.0	ND<25	ND<1.0	120	ND	NA
B32-0.5	0.5	04/16/2020	93	1,600	440	ND<1.0	ND<25	ND<1.0	1,200	ND	NA
B32-3.0	3.0	04/16/2020	ND<5.0	85	33	ND<1.0	ND<25	ND<1.0	50	ND	NA
USEPA RSLs for Soil - Residential (µg/kg) ⁽¹⁾			1,900	2,000	1,900	--	1,700	34	490	Varies	Varies
DTSC SL for Soil - Residential (µg/kg) ⁽²⁾			2,300	--	--	--	--	--	450	Varies	Varies

Key:
NA - Not analyzed
(a) = Environmental Protection Agency (EPA) Region IX Regional Screening Levels for Residential Soils
bgs = below ground surface
DTSC = California State Department of Toxic Substances Control
ESLs= Environmental Screening Levels
ND<X = not detected above the method detection limit
SL = Screening level
USEPA = United States Environmental Protection Agency
µg/kg = micrograms per kilogram

Notes:
(1) RSLs. November 2019 (Rev. 2) for soil.
(2) DTSC Human and Ecological Risk Office (HERO) Human Health Risk Assessment (HHRA) Note 3, April 2019, Cancer endpoint value
-- = USEPA RSL Not Established; or Note 3 References USEPA RSLs
Yellow highlighted results exceed RSL and SL

Table 3, Summary of Groundwater Analytical Results - Hexavalent Chromium
17631 Cameron Lane
Huntington Beach, California

Sample ID	Date	Hexavalent Chromium (µg/L)
		USEPA Method 218.6
GW-1	05/11/2020	ND<0.038
GW-2	05/11/2020	ND<0.038
GW-3	05/11/2020	ND<0.038
WS-1	05/11/2020	ND<0.038
WS-2	05/11/2020	0.086 J

Key:

J = Result is less than the reporting limit but greater than or equal to the method detection limit

µg/L = micrograms per liter

ND<X = not detected above the reporting limit or method detection limit

USEPA = United States Environmental Protection Agency

WS-1 = Duplicate

WS-2 = Equipment blank